This assignment will not be graded and is only for practice.

Key Points: 1

Let $T : V \to W$ be a linear transformation. We define kernel of T (denoted by $ker(T)$) and image of T (denoted by $Im(T)$) as follows:

$$Ker(T) = \{v \in V \mid T(v) = 0\}$$
$$Im(T) = \{w \in W \mid \text{ there exists } v \in V \text{ for which } T(v) = w\}$$

T is injective (i.e., T is a monomorphism) if and only if $Ker(T) = \{0\}$.

A linear transformation $T : V \to W$ is surjective (i.e., T is epimorphism) if and only if $Im(T) = W$.

$Nullity(T) = dim(Ker(T))$ and $Rank(T) = dim(Im(T))$.

Let $T : V \to W$ be a linear transformation. Then

$$Rank(T) + Nullity(T) = dim(V).$$

This is known as the rank nullity theorem.

Let $T : V \to W$ be a linear transformation. T is injective if and only if $Nullity(T) = 0$. T is surjective if and only if $Rank(T) = dim(W)$.

Consider the following linear transformation:

$$T : \mathbb{R}^3 \to \mathbb{R}^2$$
$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer the following questions:

1) Which of the following Options is true? **1 point**

[Hint: $T(x, y, z) = (2x + 3z, 4y + z) = (0, 0)$, i.e., $2x + 3z = 0$ and $4y + z = 0$]

- $ker(T) = \{(3y, y, -4y) \mid y \in \mathbb{R}\}$
- $ker(T) = \{(6y, y, 4y) \mid y \in \mathbb{R}\}$
- $ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$
- $ker(T) = \{(3y, y, 4y) \mid y \in \mathbb{R}\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$

2) Which of the following is a basis of $Ker(T)$? **1 point**

- $\{(3, 1, -4)\}$
- $\{(6, 0, -4), (0, 1, -4)\}$
- $\{(6, 1, -4)\}$
- $\{(3, 0, -2), (0, 1, -4)\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(6, 1, -4)\}$

3) Find the nullity of T.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

4) Find the rank of T.

[Hint: use rank nullity theorem.]

No, the answer is incorrect.
Score: 0

Feedback:

– Here dim(V) = 3 as V = R^3.
Accepted Answers:

(Type: Numeric) 2

1 point

5) Which of the following is true?

1 point

- ○ T is both injective and surjective.
- ○ T is injective but not surjective.
- ○ T is surjective but not injective.
- ○ T is neither injective nor surjective.

No, the answer is incorrect.
Score: 0

Accepted Answers:

T is surjective but not injective.

6) Which option represents the kernel and image of the following linear transformation? **1 point**

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$
$$T(x, y) = (x, 0)$$

- $ker(T) = Span\{(1,0)\}, Im(T) = Span\{(1,0)\}.$
- $ker(T) = Span\{(1,0)\}, Im(T) = Span\{(0,1)\}.$
- $ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}.$
- $ker(T) = Span\{(0,1)\}, Im(T) = Span\{(0,1)\}.$

No, the answer is incorrect.

Score: 0

Feedback:

- Ker(T) = {(0, y) | y ∈ R}.

Accepted Answers:

$ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}.$

Key Points: 2

Let $T : V \to W$ be a linear. Follow the steps below to find bases for the null space and the range space of T.

Step 1: Find the matrix A corresponding to T with respect to some standard ordered bases $\beta = \{v_1, v_2, ..., v_n\}$ and $\gamma = \{w_1, w_2, ..., w_m\}$ for V and W respectively.

Step 2: Use row reduction on A to obtain the matrix R which is in reduced row echelon form.

Step 3: The basis of the solution space of $Rx = 0$ is the basis of null space of matrix A and can be obtained by finding the pivot and non-pivot columns (dependent and independent variables) as seen earlier.

Step 4: The vectors $\begin{bmatrix} c_{11} \\ c_{12} \\ . \\ . \\ . \\ c_{1n} \end{bmatrix}, \begin{bmatrix} c_{21} \\ c_{22} \\ . \\ . \\ . \\ c_{2n} \end{bmatrix}, ..., \begin{bmatrix} c_{k1} \\ c_{k2} \\ . \\ . \\ . \\ c_{kn} \end{bmatrix}$ form a basis of the null space of A precisely when the vectors $v_1^{'}, v_2^{'}, ..., v_k^{'} \in \ker(T)$ where $v_i^{'} = \sum_{j=1}^{n} c_{ij} v_j$, form a basis for $\ker(T)$. Use the basis obtained in step 3 to thus get a basis for $\ker(T)$.

Step 5: Recall that if $i_1, i_2, \ldots, i_r$ are the columns of R containing the pivot elements, then the same columns of A form a basis for the column space of A.

Step 6: The vectors $\begin{bmatrix} d_{11} \\ d_{12} \\ . \\ . \\ . \\ d_{1m} \end{bmatrix}, \begin{bmatrix} d_{21} \\ d_{22} \\ . \\ . \\ . \\ d_{2m} \end{bmatrix}, ..., \begin{bmatrix} d_{r1} \\ d_{r2} \\ . \\ . \\ . \\ d_{rm} \end{bmatrix}$ form a basis of the column space of A precisely when the vectors $w_1^{'}, w_2^{'}, ..., w_r^{'} \in im(T)$ where $w_i^{'} = \sum_{j=1}^{m} d_{ij} w_j$, form a basis for $im(T)$. Use the basis obtained in step 5 to thus get a basis for $im(T)$.

Consider the following linear transformation:

$$T : \mathbb{R}^3 \to \mathbb{R}^3$$
$$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z)$$

Answer the following questions 7-9 using the information given above.

7) Which of the matrices corresponds the given linear transformation T with respect to the standard ordered basis of $\mathbb{R}^3$ for both the domain and the codomain? **1 point**

- $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 3 \\ 0 & 1 & 0 \end{bmatrix}$
- $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

8) What will be the kernel of T? **1 point**

- $\{(x, 0, z), (0, y, z) \mid x, y, z \in \mathbb{R}\}$.
- $\{(z, 0, z), (0, -z, z) \mid z \in \mathbb{R}\}$.
- $\{(z, -z, z) \mid z \in \mathbb{R}\}$.

- $\{(x, 0, z), (0, -x, z) \mid x, z \in \mathbb{R}\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(z, -z, z) \mid z \in \mathbb{R}\}$.

9) What will be the basis of the image space of T found by the algorithm mentioned in the Key Points ? ***1 point***

- $\{(1, 0, -1), (2, 3, 1)\}$
- $\{(1, 0, 0), (0, 1, 0)\}$
- $\{(1, 2, 0), (0, 3, 3)\}$
- $\{(-1, 1, 3)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 2, 0), (0, 3, 3)\}$

Your score is: 0/9